

Robotics

Omkar Tasgaonkar

otasgaonkar@ucla.edu

This is a set of notes covering interesting concepts I find while studying the Modern Robotics textbook. I will try to bring analytic theory whenever applicable.

Textbooks Used to Develop these Notes:

1. *Modern Robotics* - Kevin Lynch, Frank Park
2. *Calculus on Manifolds* - Michael Spivak
3. *Mathematical Analysis* - Tom Apostol

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NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Holonomic Constraints

1.1 Basic Multivariable Analysis

Note: The following notes may not be entirely rigorous! I have attempted to write them logically to the best of my abilities so that they follow a structured narrative.

We begin with the exploration of multivariate analysis. When we deal with a multivariate function in $\mathbb{R}$, we deal with some map of the form $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. We can define this more formally via vector operations as the following:

$$f \left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right) = \begin{bmatrix} f_1(x_1, x_2, \dots, x_n) \\ \vdots \\ f_m(x_1, x_2, \dots, x_n) \end{bmatrix} \quad (1)$$

such that $\forall j \in [1, m] : f_j : \mathbb{R}^n \rightarrow \mathbb{R}$. Our original definition of the derivative in $\mathbb{R}$ was:

$$f'(x) := \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

or equivalently

$$\lim_{z \rightarrow x} \frac{f(z) - f(x) - f'(x)(z - x)}{z - x} = 0$$

Using the substitution $h = z - x$, we also get the equivalent definition:

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x) - f'(x)h}{h} = 0$$

In which case, our $\epsilon - \delta$ definition would be:

$$\forall \epsilon > 0, \exists \delta > 0, \forall h \in B(0, \delta) : \frac{|f(x + h) - f(x) - f'(x)h|}{|h|} < \epsilon$$

However if we try to plainly substitute the definition for a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ into the first definition, we get "division of two vectors" which is not a defined operation for vector spaces (they are only abelian groups under addition).

We consider the following lemma in order to understand how to extend the derivative to the multivariate case:

Lemma 1.1. $f'(x)$ exists $\iff \left\{ \frac{1}{\delta} \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| \right\}_{\delta > 0} \rightarrow 0$

Proof. $\implies$: If $f'(x)$ exists, then:

$$\begin{aligned} \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \implies \forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0, \forall z \in B(x, \delta) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \implies |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot |z - x| < \epsilon \cdot \delta \end{aligned}$$

Because this is true for all z in the punctured δ -ball around x , we have an upper bound, so we take the supremum:

$$\implies \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta$$

And so we rearrange the terms to get a δ -net that converges such that:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0 : \frac{1}{\delta} \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon$$

Thus proving the forward direction.

$\Leftarrow$: Now for the converse, consider:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0 : \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta$$

Now for any arbitrary δ -ball around x where $\delta \leq \delta_0$, we have that:

$$\begin{aligned} & \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta \\ \implies & \forall z \in B(x, \delta) \setminus \{x\} : \frac{1}{\delta} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \\ \implies & \forall z \in B(x, \delta_0) \setminus B(x, \delta) : \frac{|f(z) - f(x) - f'(x)(z - x)|}{|z - x|} < \frac{1}{\delta} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \end{aligned}$$

Since δ was arbitrary as long as it was less than or equal to δ_0 , we observe:

$$\begin{aligned} & \forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0, \forall z \in B(x, \delta_0) \setminus B(x, \delta) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \iff & \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in \bigcup_{\delta \in (0, \delta_0]} (B(x, \delta_0) \setminus B(x, \delta)) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \iff & \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \bigcap_{\delta \in (0, \delta_0]} B(x, \delta) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \end{aligned}$$

Now because $\mathbb{R}$ is a T_1 space, we see that the intersection in the previous statement is equivalent to $\{x\}$, so we get:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon$$

□

We can define the function:

$$r(z) = \frac{f(z) - f(x) - f'(x)(z - x)}{z - x}$$

or $r(h)$ if we say $z = x + h$. Then taking limit as $z \rightarrow x$ (or $h \rightarrow 0$), we have that $r(z) \rightarrow 0$. Then the idea of the linear approximation is that:

$$f(z) = f(x) + f'(x)(z - x) + r(z)(z - x)$$

and the lemma above tells us that the worst remainder (similar to oscillation, which we called the "worst case net") converges to 0.

Definition 1.1. Given some $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ in the form of equation (1), then for some $i \in [1, m], j \in [1, n]$:

$$\frac{\partial f_i(x)}{\partial e_j} := \lim_{h \rightarrow 0} \frac{f_i(x + e_j h) - f_i(x)}{h}$$

where $\{e_1, \dots, e_n\}$ is the canonical basis of $\mathbb{R}^n$. This is the definition of the partial derivative.

Since each $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, we see that the partial derivative is scalar valued. We now show that it admits the linear approximation property. We can rewrite our limit definition as:

$$\lim_{h \rightarrow 0} \left(\frac{f_i(x + e_j h) - f_i(x)}{h} - \frac{\partial f_i(x)}{\partial e_j} \right) = \lim_{h \rightarrow 0} \frac{f_i(x + e_j h) - f_i(x) - \left(\frac{\partial f_i(x)}{\partial e_j} \right) h}{h} = 0$$

So with an argument similar to Lemma 1.1, we see that $f_i(z) \approx f_i(x) + \left(\frac{\partial f_i}{\partial e_j} \right) h$. Now observe that:

$$f(z) := \begin{bmatrix} f_1(z) \\ \vdots \\ f_m(z) \end{bmatrix}$$

Lemma 1.2. For all $i \in [1, m]$ and some $x \in \mathbb{R}^n$ and an e_j :

$$\frac{\partial f_i(x)}{\partial e_j} \text{ exist}$$

if and only if:

$$\frac{\partial f(x)}{\partial e_j} \text{ exists}$$

Proof. $\implies$: We claim that the partial derivative of f takes the following form:

$$\frac{\partial f(x)}{\partial e_j} = \begin{bmatrix} \frac{\partial f_1(x)}{\partial e_j} \\ \vdots \\ \frac{\partial f_m(x)}{\partial e_j} \end{bmatrix}$$

We have some δ_i given by the existence of each partial derivative, so we define $\delta_0 := \min\{\delta_i : 1 \leq i \leq m\}$. Then for all $h \in B(0, \delta_0)$:

$$\frac{\|f(x + e_j h) - f(x) - \left(\frac{\partial f(x)}{\partial e_j} \right) h\|}{|h|} \leq \frac{1}{|h|} \sum_{i=1}^m \left| f_i(x + e_j h) - f_i(x) - \left(\frac{\partial f_i(x)}{\partial e_j} \right) h \right| < m\epsilon$$

where we used the inequality that the $\|v\| \leq \sum |v_i|$.

$\Leftarrow$: We claim that:

$$\frac{\partial f_i(x)}{\partial e_j} = e_i \cdot \frac{\partial f(x)}{\partial e_j}$$

We now use the inequality $|v_i| \leq \|v\|$ and find that $\forall i \in [1, m]$:

$$\frac{|f_i(x + e_j h) - f_i(x) - (e_i \cdot \frac{\partial f(x)}{\partial e_j}) h|}{|h|} = \frac{|e_i \cdot f(x + e_j h) - e_i \cdot f_i(x) - (e_i \cdot \frac{\partial f(x)}{\partial e_j}) h|}{|h|}$$

$$= \frac{|e_i \cdot (f(x + e_j h) - f(x) - \frac{\partial f(x)}{\partial e_j} h)|}{|h|} \leq \frac{\|f(x + e_j h) - f(x) - (\frac{\partial f(x)}{\partial e_j} h)\|}{|h|} < \epsilon$$

so the same δ suffices. □

Lemma 1.3. *The partial derivative of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ exists for some e_j if and only if there exists some $r(h) \in \mathcal{M}_{m \times 1}(\mathbb{R})$ such that:*

$$f(x + h e_j) = f(x) + \frac{\partial f(x)}{\partial e_j} h + r(h) h \wedge \lim_{h \rightarrow 0} r(h) = 0$$

Proof. $\implies$: By the previous Lemma 1.2 and the results of Lemma 1.1, we can say that there exists some $r_i(h)$ for each f_i , such that:

$$f_i(x + h e_j) = f_i(x) + \frac{\partial f_i(x)}{\partial e_j} h + r_i(h) h$$

Then, we can define $r(h)$ as:

$$r(h) = \begin{bmatrix} r_1(h) \\ \vdots \\ r_m(h) \end{bmatrix}$$

which tends to 0 by definition.

$\Leftarrow$: We can rewrite the expression as:

$$r(h) = \frac{f(x + h e_j) - f(x) - \frac{\partial f(x)}{\partial e_j} h}{h}$$

so $\|r(h)\| \rightarrow 0$ implies that the quotient also tends to zero. □

Notice that the proofs did not use any properties of the basis vector e_j itself, so we can extend the definition:

Definition 1.2. *Given some $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ in the form of equation (1), then for some $v \in \mathbb{R}^n$ and $i \in [1, m], j \in [1, n]$:*

$$\frac{\partial f_i(x)}{\partial v} := \lim_{h \rightarrow 0} \frac{f_i(x + h v) - f_i(x)}{h}$$

and similarly for f . Such a derivative is called the Gateaux derivative.

We can extend the Gateaux derivative to obtain the Frechet derivative. The Gateaux derivative limited itself to a specific direction of the vector v (and then used h as the variable for the limit).

Definition 1.3. *The Frechet derivative of a function f at $x \in \mathbb{R}^n$ is:*

$$Df(x) : \lim_{v \rightarrow 0} \frac{\|f(x + v) - f(x) - (Df(x))(v)\|}{\|v\|} = 0$$

or equivalently:

$$\forall \epsilon > 0, \exists \delta > 0, \forall v \in B(0, \delta) : \frac{\|f(x + v) - f(x) - (Df(x))(v)\|}{\|v\|} < \epsilon$$

where the norm induces a metric so that $v \in B(0, \delta)$ (in this case 0 is the 0 vector), is the same as saying $\|v\| \in B(0, \delta)$ or equivalently $\|v\| < \delta$.

The form of the limit is similar to the idea of linear approximation. Since $v \in \mathbb{R}^n$, then we must have that $Df(x) \in \mathcal{M}_{m \times n}(\mathbb{R})$. If the function is Frechet differentiable, then $Df(x)$ is the Jacobian:

$$Df(x) = \begin{bmatrix} \frac{\partial f_1(x)}{\partial e_1} & \dots & \frac{\partial f_1(x)}{\partial e_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(x)}{\partial e_1} & \dots & \frac{\partial f_m(x)}{\partial e_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial f(x)}{\partial e_1} & \dots & \frac{\partial f(x)}{\partial e_n} \end{bmatrix}$$

Proof. If we chose the family of vectors $\{he_j : h \in \mathbb{R} \wedge j \in [1, n]\}$, then differentiability tells us:

$$\forall \epsilon > 0, \exists \delta > 0, \forall h \in B(0, \delta) : \frac{\|f(x + he_j) - f(x) - h(Df(x))e_j\|}{|h|} < \epsilon$$

telling us that:

$$\frac{\partial f(x)}{\partial e_j} = Df(x)e_j$$

which is the j -th column of the Jacobian matrix. □

Notice that Lemma 1.2 was a statement on equivalence; however, the converse is not necessarily true for the Jacobian matrix. The total differentiability of f at x gives us the existence of partial derivatives, but not necessarily the converse. In order to prove this, we require the following concept:

1.1.1 MVT in the Multivariable Case

Theorem 1.1. *Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous on the closed interval $[a, b]$ and differentiable on (a, b) , then:*

$$\exists z \in (a, b) : f'(z) = \frac{f(b) - f(a)}{b - a}$$

Proof. Since f is continuous on a compact interval, it achieves the extrema by EVT. Now consider the function:

$$h(x) = \frac{f(b) - f(a)}{b - a}(x - a) - f(x) + f(a)$$

which at both $x = a$ and $x = b$, we have that $h(x) = 0$, so by Rolle's Theorem, there exists some extrema within the interval, so that:

$$\exists z \in (a, b) : h'(z) = 0 \implies f'(z) = \frac{f(b) - f(a)}{b - a}$$

□

If we try to apply this proof to a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, then in order to apply Theorem 1.1, we require some compact interval given f is continuous.

Now, the following lemma will be useful in exploring MVT:

Lemma 1.4. *Any open, convex subset $U \subseteq \mathbb{R}^n$ is path connected.*

Proof. Openness does not play an important role here. We can simply take the continuous function:

$$f(\alpha) = \alpha x + (1 - \alpha)y$$

for some fixed $x, y \in U$. By convexity, we know $\text{Ran}(f) \subseteq U$. □

Lemma 1.5. *An open ball $B(x, r)$ is convex.*

Proof. Take $y, z \in B(x, r)$, then:

$$|\alpha y + (1 - \alpha)z - x| = |\alpha(y - x) + (1 - \alpha)(z - x)| \leq \alpha r + (1 - \alpha)r = r$$

□

Theorem 1.2. *Given $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, we have that for some open, convex subset $U \subseteq \mathbb{R}^n$ (on which f_i is differentiable) and given two points a, b :*

$$\exists z \in U : Df_i(z)(b - a) = f_i(b) - f_i(a)$$

where $Df_i(z)$ is the gradient vector at z .

Proof. Since U is convex, we can take the function for $t \in [0, 1]$:

$$h(t) = tb + (1 - t)a$$

such that $h(0) = a$ and $h(1) = b$. Then applying Theorem 1.1 to the function $f_i \circ h$ (which too is continuous on $[0, 1]$ and differentiable on $(0, 1)$), we see that:

$$\exists c \in (0, 1) : (f_i \circ h)'(c) = \frac{f_i(h(1)) - f_i(h(0))}{1 - 0} = f_i(b) - f_i(a)$$

and $(f_i \circ h)'(c) = Df_i(h(c)) \cdot h'(c) = Df_i(c)(b - a)$, where:

$$Df_i(x) = \left[\frac{\partial f_i(x)}{\partial e_1} \quad \dots \quad \frac{\partial f_i(x)}{\partial e_n} \right]$$

We used the multivariate chain rule in this proof, where $(f \circ h)'(t) = Df(h(t))Dh(t)$. In this case, we defined:

$$h(t) = \begin{bmatrix} tb_1 + (1 - t)a_1 \\ \vdots \\ tb_n + (1 - t)a_n \end{bmatrix}$$

and since each $h_i : \mathbb{R} \rightarrow \mathbb{R}$, then $Dh(t)$ is an $n \times 1$ matrix in the e_1/t direction, the form of which is given by Lemma 1.2:

$$Dh(t) = \begin{bmatrix} b_1 - a_1 \\ \vdots \\ b_n - a_n \end{bmatrix} = b - a$$

□

Corollary 1.1. *If $\exists k \in \mathbb{R}, \exists j \in [1, n] : b - a = ke_j$ and the partial derivative of f_i exists in the e_j direction, then:*

$$\exists z \in \text{span}(e_j) : \frac{\partial f_i(z)}{\partial e_j} k = f_i(b) - f_i(a)$$

Notice this corollary does not require full differentiability of f_i (which means it does not require the gradient to be the total derivative of f at said point).

Lemma 1.6. *The function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is totally differentiable at a point if:*

1. *The partial derivatives exist*
2. *The partial derivatives are continuous at that point*

Proof. We want to show:

$$\forall \epsilon > 0, \exists \delta > 0, \forall v \in B(0, \delta) : \frac{\|f(x+v) - f(x) - (Df(x))v\|}{\|v\|} < \epsilon$$

The expression $(Df(x))(v)$ gives us the following vector:

$$(Df(x))(v) = \begin{bmatrix} \sum_{i=1}^n \frac{\partial f_1(x)}{\partial e_i} v_i \\ \vdots \\ \sum_{i=1}^n \frac{\partial f_m(x)}{\partial e_i} v_i \end{bmatrix}$$

and we see that:

$$\frac{\|f(x+v) - f(x) - (Df(x))(v)\|}{\|v\|} \leq \frac{1}{\|v\|} \sum_{j=1}^m \left| f_j(x+v) - f_j(x) - \sum_{i=1}^n \frac{\partial f_j(x)}{\partial e_i} v_i \right|$$

Continuity of the partial derivatives give us:

$$\forall \epsilon > 0, \exists \delta_{ji} > 0, \forall v \in B(0, \delta_{ji}) : \left| \frac{\partial f_j(x+v)}{\partial e_i} - \frac{\partial f_j(x)}{\partial e_i} \right| < \epsilon$$

We set $\delta := \min\{\delta_{ji} : 1 \leq j \leq m \wedge 1 \leq i \leq n\}$. Then given some $v \in B(0, \delta)$, we apply the idea in Corollary 1.1 and see that:

$$f_j(x+v) - f_j(x) = \sum_{i=1}^n f_j(x + V_i) - f_j(x + V_{i-1})$$

where we define $V_i := v_1 e_1 + \dots + v_i e_i$. Then, we can repeatedly apply Corollary 1.1 and get:

$$= \frac{1}{\|v\|} \sum_{j=1}^m \left| \sum_{i=1}^n \left(\frac{\partial f_j(z_i)}{\partial e_i} - \frac{\partial f_j(x)}{\partial e_i} \right) v_i \right| \leq \frac{\epsilon \cdot mn \cdot \|v\|}{\|v\|} = \epsilon \cdot mn$$

□

The MVT introduced in Theorem 1.2 fails for vector valued functions as we cannot guarantee the same z satisfies the equality for all f_i ; however, there are some generalizations to this case which we can expound upon later. Finally, we can expand the idea of linear approximation to the total vector valued function:

Lemma 1.7. *$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is totally differentiable if and only if there exists $r(v) : \mathbb{R}^n \rightarrow \mathcal{M}_{m \times n}(\mathbb{R})$ such that:*

$$f(x+v) = f(x) + Df(x)(v) + r(v)v \wedge \lim_{v \rightarrow 0} \|r(v)\| = 0$$

Proof. Observe that we use the matrix norm, which is defined as:

$$\|A\| = \sup_{x \in \mathbb{R}^n} \frac{\|Ax\|}{\|x\|}$$

$$\forall x \in \mathbb{R}^n : \frac{\|Ax\|}{\|x\|} = \sqrt{\frac{\sum_{i=1}^m (\sum_{j=1}^n A_{ij} x_j)^2}{\sum_{j=1}^n x_j^2}}$$

$$\leq \sqrt{\frac{\sum_{i=1}^m (\sum_{j=1}^n A_{ij}^2 \sum_{j=1}^n x_j^2)}{\sum_{j=1}^n x_j^2}} = \sqrt{\sum_{i=1}^m \left(\sum_{j=1}^n A_{ij}^2 \right)}$$

and since this is true for all $x \in \mathbb{R}^n$, we have a concrete upper bound, so:

$$\|A\| \leq \sqrt{\sum_{i=1}^m \left(\sum_{j=1}^n A_{ij}^2 \right)}$$

Now, we can use the fact that $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ and that its total derivative exists as seen by:

$$\frac{|f_i(x+v) - f_i(x) - Df_i(x)(v)|}{\|v\|} = \frac{|e_i^T(f(x+v) - f(x) - Df(x)(v))|}{\|v\|} \leq \frac{\|f(x+v) - f(x) - Df(x)(v)\|}{\|v\|}$$

Then, we say that equivalently there exists some $r_i(v) : \mathbb{R}^n \rightarrow \mathcal{M}_{1 \times n}$ such that:

$$r_i(v)v = f_i(x+v) - f_i(x) - Df_i(x)(v)$$

such that:

$$\begin{aligned} r_i(v) &= \left[(f_i(x+v) - f_i(x) - Df_i(x)(v)) \frac{v_1}{\|v\|^2} \quad \cdots \quad (f_i(x+v) - f_i(x) - Df_i(x)(v)) \frac{v_n}{\|v\|^2} \right] \\ \implies \|r_i(v)\| &= \|f_i(x+v) - f_i(x) - Df_i(x)(v)\| \sqrt{\frac{1}{\|v\|^2}} = \frac{\|f_i(x+v) - f_i(x) - Df_i(x)(v)\|}{\|v\|} \rightarrow 0 \end{aligned}$$

The converse follows from the fact that $\|Ax\| \leq \|A\| \cdot \|x\|$. Now, we can define:

$$r(v) := \begin{bmatrix} r_1(v) \\ \vdots \\ r_m(v) \end{bmatrix}$$

for which by the inequality for matrix norm we derived above, we have that:

$$\|r(v)\| \leq \sqrt{\sum_{i=1}^m (f_i(x+v) - f_i(x) - Df_i(x)(v))^2} \cdot \frac{1}{\|v\|^2} = \frac{1}{\|v\|} \cdot \|f(x+v) - f(x) - Df(x)(v)\| \rightarrow 0$$

□

1.2 Inverse and Implicit Function Theorems

The idea of the inverse function theorem stems from our idea of the linear approximation where $f(x+v) \approx f(x) + Df(x)v$. Then $v \approx Df(x)^{-1}(f(x+v) - f(x))$.

1.3 Configuration Spaces

The idea of configuration spaces is intertwined with the idea of manifolds, which I have not studied, so for this section we will limit ourselves to Euclidean configuration spaces.